Robust Online POMDP

1 Contributions

the main contributions this research provides are:

1. A formulation of robust online POMDP planning under simultaneous transition and observation uncertainty.
2. A novel definition of projected uncertainty sets over sampled supports, enabling robust reasoning on partial online search trees, and guarantee consistency with an extendable full model.
3. A formal support-based representation of history nodes for online planning in sampled state and observation spaces.
4. We derive an explicit decomposition of the error between the full value function and the projected value function into two sources: inside-support mismatch and omitted trajectories contribution.
5. a deterministic finite-horizon non-robust error bound between the theoretical and projected value functions, showing how the error accumulates through belief mismatch, model mismatch on the sampled support, and probability mass outside the sampled support.
6. a **deterministic** robust error bound in which the inside-support term is controlled through a worst-case kernel-level mismatch and the omitted term is controlled through worst-case leakage outside the sampled support.
7. A tractable rectangular robust formulation that motivates future robust online planning algorithms.

2 Uncertainty Set

currently we consider uncertainty both in the transition model P_T and in the observation model P_Z. We will denote the nominal models as P_T^o, P_Z^o. Given some distance metric $D(\cdot||\cdot)$ we define the uncertainty sets over the models:

$$\mathcal{P}_T(s,a) = \left\{P_T(\cdot|s,a) : \forall s' \in S,\ D\left(P_T(s'|s,a)||P_T^o(s'|s,a)\right) \leqslant \rho(s,a)\right\} \tag{1}$$

$$\mathcal{P}_Z(s) = \left\{P_Z(\cdot|s) : \forall z \in Z,\ D\left(P_Z(z|s)||P_Z^o(z|s)\right) \leqslant \rho(s)\right\} \tag{2}$$

Later i need to explain the definition of $\rho(s,a)$ and $\rho(s)$

we introduce a new concept we call projected uncertainty sets. during online planning, compute the value function of history nodes, who are composed of sampled states and observations, and only see a subset of the spaces. we can only evaluate on what the agent actually sees/samples. the projected uncertainty is defines a set of valid sub-probability vectors such that they can be extended into a full valid distribution that lies within the original uncertainty set, and therefore we are guaranteed that the values we compute represent the true worst case in the of the current tree state with respect to the problem error radius. we denote with a $\hat{\mathcal{P}}$ the projected sets.

$$\widehat{\mathcal{P}}_T(s;ha) \ = \left\{ P_T^{\text{in}} \in \mathbb{R}_+^{|S^{\text{in}}|} : \exists \ P_T^{\text{out}} \in \mathbb{R}_+^{|S^{\text{out}}|} \ s.t. \ \left(P_T^{\text{in}}, P_T^{\text{out}}\right) \in \mathcal{P}_T \right\} \tag{3}$$

$$\widehat{\mathcal{P}}_Z(s;ha) \ = \left\{ P_Z^{\text{in}} \in \mathbb{R}_+^{|Z^{\text{in}}|} : \exists \ P_Z^{\text{out}} \in \mathbb{R}_+^{Z^{\text{out}}|} \ s.t. \ \left(P_Z^{\text{in}}, P_Z^{\text{out}}\right) \in \mathcal{P}_Z \right\} \tag{4}$$

important to note that P_T^{in}, P_Z^{in} are sub-probability vectors and do not correspond to a valid distribution.
the history nodes ha store the meta data regarding the current state of that node, meaning all the state and observations seen so far. more specifically, for a history node $h^k a$: [switch to ha?]

$$h^k a = \left\{ s_i^k \right\}_{i=0}^{N\left(h^k a\right)} \cup \left\{ z_i^k \right\}_{i=0}^{N\left(h^k a\right)}$$

although the form above allows duplicated samples, it still correctly defines the support over the spaces:

$$\text{supp}_S\left(h^k a\right) = \left\{ s \in S : \ s \in \left\{ s_i^k \right\}_{i=0}^{N\left(h^k a\right)} \right\}$$

$$\text{supp}_Z\left(h^k a\right) = \left\{ z \in Z : \ z \in \left\{ z_i^k \right\}_{i=0}^{N\left(h^k a\right)} \right\}$$

additional definitions: S^{in} and Z^{in} correspond to state/observation samples that were sampled during planning. [should be a function of $h^k a$]

$$S = S^{\text{in}} \cup S^{\text{out}} \quad s.t \quad S^{\text{in}} = \text{supp}_S(h^k a)$$

$$Z = Z^{\text{in}} \cup Z^{\text{out}} \quad s.t \quad Z^{\text{in}} = \text{supp}_Z(h^k a)$$

3 value function

in this section we want to explicitly derive the value function, to later be used for the bounds. considering horizon H, the return at time step t is defined as:

$$G_t = \sum_{k=t}^{t+H} r(s_k, a_k) \tag{5}$$

and the value function is defined as: [the return is generally defined as $G_t = \sum_{k=t}^{t+H} r(b_k, a_k)$; it is true that for state-dependent rewards we get to (15); define more formally what is h_t (if it is a sequence of actions-observations, then the reward should be still belief-dependent - pls verify);

clarify how is the action determined as a function of the policy]

$$V_t^\pi(h_t) = \mathbb{E}^\pi[G_t|h_t, b_t] = \mathbb{E}^\pi\left[r(s_t, a_t) + \sum_{k=t+1}^{t+H} r(s_k, a_k)|h_t, b_t\right] = \tag{6}$$

$$\mathbb{E}^\pi[r(s_t, a_t)|h_t, b_t] + \mathbb{E}^\pi\left[\sum_{k=t+1}^{t+H} r(s_k, a_k)|h_t, b_t\right] = \tag{7}$$

$$\mathbb{E}_{s_t\sim b(\cdot|h_t)}\left[r(s_t, a_t) + \mathbb{E}_{h_{t+1:t+H}\sim p(\cdot|h_t,\ a_t, s_t)}\left[G_{t+1}|h_t, b_t\right]\right] \tag{8}$$

we first show how to define the history distribution $p(\cdot|h_t,\ a_t, s_t)$ with the models: [should condition on the policy sequence?]

$$p(h_{t+1:t+H}|h_t,\ a_t, s_t) = p(a_{t+1:t+H},\ z_{t+1:t+H}|a_{0:t},\ z_{1:t},\ s_t) \overbrace{=}^{marginalize\ over\ states} \tag{9}$$

$$\sum_{s_{t+1:t+H}} p(a_{t+1:t+H},\ z_{t+1:t+H}, s_{t+1:t+H}|a_{0:t},\ z_{1:t},\ s_t) \overbrace{=}^{chain\ rule\ (states)} \tag{10}$$

$$\sum_{s_{t+1:t+H}} p(a_{t+1:t+H},\ z_{t+1:t+H}|a_{0:t},\ z_{1:t},\ s_{t:t+H}) \cdot p(s_{t+1:t+H}|a_{0:t},\ z_{1:t}, s_t) \overbrace{=}^{chain\ rule\ (observations)} \tag{11}$$

$$\sum_{s_{t+1:t+H}} p(a_{t+1:t+H}|a_{0:t},\ z_{1:t+H},\ s_{t:t+H}) \cdot p(s_{t+1:t+H}|a_{0:t},\ z_{1:t}, s_t) \cdot p(z_{t+1:t+H}|a_{0:t},\ z_{1:t},\ s_{t:t+H}) \overbrace{=}^{markovian} \tag{12}$$

$$\sum_{s_{t+1:t+H}} \prod_{k=t}^{t+H-1} p(a_{k+1}|h_{k+1}) \cdot p(s_{k+1}|a_k, s_k) \cdot p(z_{k+1}|s_{k+1}) \overbrace{=}^{writing\ with\ models\ notations} \tag{13}$$

$$\sum_{s_{t+1:t+H}} \prod_{k=t}^{t+H-1} \pi(a_{k+1}|h_{k+1}) \cdot P_T(s_{k+1}|a_k, s_k) \cdot P_Z(z_{k+1}|s_{k+1}) \tag{14}$$

now we want to plug (14) in (8):[policy should be time-indexed (since it's finite horizon)]

$$V_t^\pi(h_t) = \underbrace{\sum_{a_t} \pi(a_t \mid h_t)}_{\text{only if stochastic}} \Bigg[\sum_{s_t\in S} b(s_t \mid h_t) \Bigg[r(s_t, a_t) + \sum_{a_{t+1:t+H}} \sum_{z_{t+1:t+H}} \sum_{s_{t+1:t+H}} \prod_{k=t}^{t+H-1} \underbrace{\pi(a_{k+1} \mid h_{k+1})}_{\text{only if stochastic}} P_T(s_{k+1} \mid a_k, s_k)\, P_Z(z_{k+1} \mid s_{k+1}) \left[\sum_{j=t+1}^{t+H} r(s_j, a_j)\right]\Bigg]\Bigg] \tag{15}$$

we consider a deterministic policy (the tree policy), and therefore we can remove the policy out of the equation: [(i) for deterministic policies, should replace a_j with $\pi_j(h_j)$; (ii) if we later consider MCTS, it calculates a stochastic policy]

$$
\begin{aligned}
V_t^{\pi}(h_t) &= \sum_{s_t \in S} b(s_t \mid h_t) \Bigg[r(s_t, a_t) + \sum_{z_{t+1:t+H}} \sum_{s_{t+1:t+H}} \prod_{k=t}^{t+H-1} P_T(s_{k+1} \mid a_k, s_k)\, P_Z(z_{k+1} \mid s_{k+1}) \Bigg[\sum_{j=t+1}^{t+H} r(s_j, a_j) \Bigg] \Bigg] \\
&= \sum_{j=t}^{t+H} \sum_{z_{t+1:j}} \sum_{s_{t:j}} \prod_{k=t}^{j-1} P_T(s_{k+1} \mid \pi_k, s_k)\, P_Z(z_{k+1} \mid s_{k+1})\, b(s_t \mid h_t)\, r(s_j, a_j)
\end{aligned} \tag{16}
$$

we consider (16) as the theoretical value function. similarly the simplified value function given that we only see a subset (inside support) of the space is define as:

$$
\widehat{V}_t^{\pi}(h_t) = \sum_{j=t}^{t+H} \sum_{z_{t+1:j} \in Z_{t+1:j}^{\text{in}}} \sum_{s_{t:j} \in S_{t:j}^{\text{in}}} \prod_{k=t}^{j-1} P_T^{\text{in}}(s_{k+1} | \pi_k, s_k)\ P_Z^{\text{in}}(z_{k+1} | s_{k+1})\ b^{in}(s_t | h_t)\ r(s_j, a_j) \tag{17}
$$

4 bounds non robust

in this section we want to derive **deterministic** bounds between the theoretical value function (16) and the simplified (projected) value function (17). before we show the bounds derivation, we split the bounds into 2 terms:

$$
\left| V_t^{\pi}(h_t) - \widehat{V}_t^{\pi}(h_t) \right| \leq \underbrace{\text{(inside mismatch)}}_{\text{when } P_{in} \neq P_{in}^o} + \underbrace{\text{(omitted trajectories contribution)}}_{\text{always}} \tag{18}
$$

for intuition, the first part refers to the mismatch over the support between the nominal kernels P_T^o, P_Z^o and the simplified kernels $P_T^{\text{in}}, P_Z^{\text{in}}$. later when we move to robust bounds, the support mismatch will be between the worst-case kernels in $\mathcal{P}_T/\widehat{\mathcal{P}}_T, \mathcal{P}_Z/\widehat{\mathcal{P}}_Z$. the second part is the density of the nominal kernels over the outside support weighting the values over that space, and again when we will move to robust bounds it will be the the outside support density of the worst-case kernels in $\mathcal{P}_T, \mathcal{P}_Z$. first we show what each term looks like, currently only looking at a single reward contribution, which will later be generalized to the full horizon. we define a single reward contribution as $U_j^{\pi}(h_t)$, such that:

$$
V_t^{\pi}(h_t) = \sum_{j=t}^{t+H} U_j^{\pi}(h_t) \tag{19}
$$

$$
\widehat{V}_t^{\pi}(h_t) = \sum_{j=t}^{t+H} \widehat{U}_j^{\pi}(h_t) \tag{20}
$$

$$
\begin{aligned}
\left| U_j^{\pi}(h_t) - \widehat{U}_j^{\pi}(h_t) \right| = \Bigg| &\sum_{z_{t+1:j}} \sum_{s_{t:j}} \prod_{k=t}^{j-1} P_T(s_{k+1} \mid \pi_k, s_k)\ P_Z(z_{k+1} \mid s_{k+1})\ b(s_t \mid h_t)\ r(s_j, a_j) \\
&- \sum_{z_{t+1:j} \in Z_{t+1:j}^{\text{in}}} \sum_{s_{t:j} \in S_{t:j}^{\text{in}}} \prod_{k=t}^{j-1} P_T^{\text{in}}(s_{k+1} \mid \pi_k, s_k)\ P_Z^{\text{in}}(z_{k+1} \mid s_{k+1})\ b^{\text{in}}(s_t \mid h_t)\ r(s_j, a_j) \Bigg|
\end{aligned} \tag{21}
$$

we now split the theoretical part into in/out support and we get our 2 terms in (18)

$$\left|U_j^{\pi}(h_t)-\widehat{U}_j^{\pi}(h_t)\right| = \Bigg|\sum_{z_{t+1:j}\in Z^{\text{in}}_{t+1:j}}\ \sum_{s_{t:j}\in S^{\text{in}}_{t:j}}\Bigg[b(s_t\mid h_t)\prod_{k=t}^{j-1}P_T(s_{k+1}\mid \pi_k,s_k)\ P_Z(z_{k+1}\mid s_{k+1})\ r(s_j,a_j)$$
$$- b^{\text{in}}(s_t\mid h_t)\prod_{k=t}^{j-1}P_T^{\text{in}}(s_{k+1}\mid \pi_k,s_k)\ P_Z^{\text{in}}(z_{k+1}\mid s_{k+1})\ r(s_j,a_j)\Bigg]$$
$$+\sum_{\substack{z_{t+1:j}\notin Z^{\text{in}}_{t+1:j}\\ \cup\\ s_{t:j}\notin S^{\text{in}}_{t:j}}}\prod_{k=t}^{j-1}P_T(s_{k+1}\mid \pi_k,s_k)\,P_Z(z_{k+1}\mid s_{k+1})\ b(s_t\mid h_t)\ r(s_j,a_j)\Bigg|\ \underbrace{\leq}_{\substack{\text{triangle}\\ \text{inequality}}} \tag{22}$$

$$\Bigg|\sum_{z_{t+1:j}\in Z^{\text{in}}_{t+1:j}}\ \sum_{s_{t:j}\in S^{\text{in}}_{t:j}}\Bigg[b(s_t\mid h_t)\prod_{k=t}^{j-1}P_T(s_{k+1}\mid \pi_k,s_k)\ P_Z(z_{k+1}\mid s_{k+1})\ r(s_j,a_j)$$
$$- b^{\text{in}}(s_t\mid h_t)\prod_{k=t}^{j-1}P_T^{\text{in}}(s_{k+1}\mid \pi_k,s_k)\ P_Z^{\text{in}}(z_{k+1}\mid s_{k+1})\ r(s_j,a_j)\Bigg]\Bigg| \tag{23}$$
$$+\Bigg|\sum_{\substack{z_{t+1:j}\notin Z^{\text{in}}_{t+1:j}\\ \cup\\ s_{t:j}\notin S^{\text{in}}_{t:j}}}\prod_{k=t}^{j-1}P_T(s_{k+1}\mid \pi_k,s_k)\ P_Z(z_{k+1}\mid s_{k+1})\ b(s_t\mid h_t)\ r(s_j,a_j)\Bigg| \tag{24}$$

we can now refer to the inside mismatch term as (23) and to the omitted trajectories contribution term as (24). for any j, $r(s_j,a_j)\in[r_{min},r_{max}]$. we define $R_{max}=\max[|r_{\min}|,\ |r_{\max}|]$. we first deal with the mismatch term (23). using the triangle inequality twice:

$$R_{max}\cdot\sum_{z_{t+1:j}\in Z^{\text{in}}_{t+1:j}}\ \sum_{s_{t:j}\in S^{\text{in}}_{t:j}}\Bigg|\prod_{k=t}^{j-1}P_T(s_{k+1}\mid \pi_k,s_k)\ P_Z(z_{k+1}\mid s_{k+1})\ b(s_t\mid h_t)$$
$$-\prod_{k=t}^{j-1}P_T^{\text{in}}(s_{k+1}\mid \pi_k,s_k)\ P_Z^{\text{in}}(z_{k+1}\mid s_{k+1})\ b^{\text{in}}(s_t\mid h_t)\Bigg| \tag{25}$$

$$\mu(\tau_{t:j}) = b(\cdot)\prod_{k=t}^{j-1}P_T(s_{k+1}\mid \pi_k,s_k)\ P_Z(z_{k+1}\mid s_{k+1}) = b(\cdot)\prod_{k=t}^{j-1}K(z_{k+1},s_{k+1}\mid s_k,\pi_k) \tag{26}$$
$$\widehat{\mu}(\tau_{t:j}) = b^{\text{in}}(\cdot)\prod_{k=t}^{j-1}P_T^{\text{in}}(s_{k+1}\mid \pi_k,s_k)\ P_Z^{\text{in}}(z_{k+1}\mid s_{k+1}) = b^{\text{in}}(\cdot)\prod_{k=t}^{j-1}\widehat{K}(z_{k+1},s_{k+1}\mid s_k,\pi_k) \tag{27}$$
$$\text{s.t.}\qquad \tau_{t:j}=\{s_t,z_{t+1},s_{t+1},\dots,z_j,s_j\}\in S_j^{\text{in}}\times Z_j^{\text{in}} \tag{28}$$
$$K(z,s'\mid s,\pi)=P_T(s'\mid \pi,s)\,P_Z(z\mid s') \tag{29}$$
$$\widehat{K}(z,s'\mid s,\pi)=P_T^{\text{in}}(s'\mid \pi,s)\,P_Z^{\text{in}}(z\mid s') \tag{30}$$

then (25) can be rewritten as:

$$R_{max} \cdot \sum_{\tau_{t:j}} |\mu(\tau_{t:j}) - \widehat{\mu}(\tau_{t:j})| = R_{max} \cdot \|\mu(\tau_{t:j}) - \widehat{\mu}(\tau_{t:j})\|_1 \tag{31}$$

R_{max} is fixed, all we need is to bound the L1 norm. we define the joint distribution up to time k as the weights $W_k, \widehat{W}_k$:

$$W_{k+1} = W_k\, K \tag{32}$$

$$\widehat{W}_{k+1} = \widehat{W}_k\, \widehat{K} \tag{33}$$

we define the distance D_k as:

$$D_k = \left\| W_k - \widehat{W}_k \right\|_1 \tag{34}$$

using the above definitions:

$$D_{k+1} = \left\| W_{k+1} - \widehat{W}_{k+1} \right\|_1 = \left\| W_k \cdot K - \widehat{W}_k \cdot \widehat{K} \right\|_1 \tag{35}$$

$$= \left\| W_k \cdot K - \widehat{W}_k \cdot K + \widehat{W}_k \cdot K - \widehat{W}_k \cdot \widehat{K} \right\|_1 \tag{36}$$

$$\underbrace{\leq}_{\substack{\text{triangle} \\ \text{inequality}}} \left\| W_k \cdot K - \widehat{W}_k \cdot K \right\|_1 + \left\| \widehat{W}_k \cdot K - \widehat{W}_k \cdot \widehat{K} \right\|_1 \tag{37}$$

$$\leq \underbrace{\|K\|_1}_{\leq 1} \cdot \left\| W_k - \widehat{W}_k \right\|_1 + \underbrace{\left\| \widehat{W}_k \right\|_1}_{\leq 1} \cdot \left\| K - \widehat{K} \right\|_1 . \tag{38}$$

by definition:

$$\|K(z, s' \mid s, \pi)\|_1 \leq \sup_s \left[\sum_z \sum_{s'} K(z, s' \mid s, \pi) \right] \tag{39}$$

$$= \sup_s \left[\sum_{s'} P_T(s' \mid \pi, s) \sum_z P_Z(z \mid s') \right] \tag{40}$$

$$\leq \sup_s \left[\sum_{s'} P_T(s' \mid \pi, s) \right] \leq 1 \tag{41}$$

the last 2 transitions comes from the fact that since both P_T and P_Z are a distribution over subset of the space the sum can be at most 1. same thing for $\|\widehat{W}_k\|_1$. therefore from (38) we get:

$$\left\| W_k - \widehat{W}_k \right\|_1 + \left\| K - \widehat{K} \right\|_1 = D_k + \left\| K - \widehat{K} \right\|_1 \tag{42}$$

now we bound $\left\| K - \widehat{K} \right\|_1$:

$$
\begin{aligned}
\|K-\widehat{K}\|_1 &= \sum_z \sum_{s'} \left|P_T \cdot P_Z - P_T^{\text{in}} \cdot P_Z^{\text{in}}\right| && (43)\\
&= \sum_z \sum_{s'} \left|P_T \cdot P_Z - P_T^{\text{in}} \cdot P_Z + P_T^{\text{in}} \cdot P_Z - P_T^{\text{in}} \cdot P_Z^{\text{in}}\right| && (44)\\
&\underbrace{\leq}_{\substack{\text{triangle}\\ \text{inequality}}} \sum_z \sum_{s'} \left|P_T \cdot P_Z - P_T^{\text{in}} \cdot P_Z\right| + \sum_z \sum_{s'} \left|P_T^{\text{in}} \cdot P_Z - P_T^{\text{in}} \cdot P_Z^{\text{in}}\right| && (45)\\
&= \sum_z \sum_{s'} P_Z \cdot \left|P_T - P_T^{\text{in}}\right| + P_T^{\text{in}} \cdot \left|P_Z - P_Z^{\text{in}}\right| && (46)\\
&= \sum_{s'} \left|P_T - P_T^{\text{in}}\right| \sum_z P_Z + \sum_{s'} P_T^{\text{in}} \sum_z \left|P_Z - P_Z^{\text{in}}\right| && (47)
\end{aligned}
$$

we can actually stop here. the kernels bound is the mismatch between both the observation models and the transition models weighted by the inner support. for convinience we denote $w_Z = \sum_{z \in Z^{\text{in}}} P_Z$, $w_T = \sum_{s' \in S^{\text{in}}} P_T^{\text{in}}$ (note that we can also use $w_Z = \sum_{z \in Z^{\text{in}}} P_Z^{\text{in}}$, $w_T = \sum_{s' \in S^{\text{in}}} P_T$, which ever one gives a tighter bound).
we can also get a looser but simpler bound by pulling the observation mismatch out of the sum over next states by taking the supremum, and the fact that the weights w_Z, w_T are at most 1, and get:

$$
\leq \|P_T - P_T^{\text{in}}\|_1 + \sup_{s'} \|P_Z - P_Z^{\text{in}}\|_1 \tag{48}
$$

let us define the bound we got with a new notation:

$$
\begin{aligned}
\Delta_T(h_t, a_t) &= \|P_T - P_T^{\text{in}}\|_1 && (49)\\
\Delta_T^*(h_t, a_t) &= w_Z \cdot \|P_T - P_T^{\text{in}}\|_1 && (50)
\end{aligned}
$$

$$
\begin{aligned}
\Delta_Z(h_t, a_t) &= \sup_{s'} \|P_Z - P_Z^{\text{in}}\|_1 && (51)\\
\Delta_Z^*(h_t, a_t) &= w_T \cdot \|P_Z - P_Z^{\text{in}}\|_1 && (52)
\end{aligned}
$$

we got that (35) $\leq$ (42), and (42) $\leq \Delta_T^*(h_t, a_t) + \Delta_Z^*(h_t, a_t) \leq \Delta_T(h_t, a_t) + \Delta_Z(h_t, a_t)$, meaning:

$$
D_{k+1} \leq D_k + \Delta_T(h_k, a_k) + \Delta_Z(h_k, a_k) \tag{53}
$$

to bound a single time step D_j we unroll from t to j:

$$
D_j \leq D_t + \sum_{k=t}^{j-1} \left(\Delta_T(h_k, a_k) + \Delta_Z(h_k, a_k)\right) \tag{54}
$$

for the initial timestep t we get $D_t = \left\| b_t - \widehat{b}_t \right\|_1 = \Delta_b$:

$$\|\mu(\tau_t) - \widehat{\mu}(\tau_t)\|_1 = \|\mu(s_t) - \widehat{\mu}(s_t)\|_1 = \left\| b(s_t \mid h_t) - \widehat{b}(s_t \mid h_t) \right\|_1 \tag{55}$$

$$D_j \le \Delta_b + \sum_{k=t}^{j-1} (\Delta_T(h_k, a_k) + \Delta_Z(h_k, a_k)) \tag{56}$$

putting it all together, meaning plugging (56) inside (31), the inside-mismatch is bounded by:

$$R_{\max} \left(\Delta_b + \sum_{k=t}^{j-1} \big(\Delta_T(h_k, a_k) + \Delta_Z(h_k, a_k) \big) \right) \tag{57}$$

we now have to bound the omitted trajectories contribution (the second term in (18)). we defined that term in (24). For simplicity of notation, we define the set of trajectories that lie in the support as:

$$\tau^{\text{in}} = \{s_t, z_{t+1}, s_{t+1}, \ldots, z_j, s_j\} \in X^{\text{in}} = S^{\text{in}} \times Z^{\text{in}},$$

and the set of trajectories that violate the support as:

$$\tau^{\text{out}} \in X^{\text{out}} = X \setminus X^{\text{in}}$$

The omitted trajectories bound derivation:

$$\left| \sum_{\tau \in X^{\text{out}}} \prod_{k=t}^{j-1} P_T(s_{k+1} \mid \pi_k, s_k)\, P_Z(z_{k+1} \mid s_{k+1})\, b(s_t \mid h_t)\, r(s_j, a_j) \right| \le \tag{58}$$

$$R_{\max} \sum_{\tau \in X^{\text{out}}} b(s_t \mid h_t) \prod_{k=t}^{j-1} K(z_{k+1}, s_{k+1} \mid \pi_k, s_k) \tag{59}$$

The transition is due to the triangle inequality and the fact that the kernel is non-negative. The total trajectory probability mass sums to 1:

$$\sum_{\tau \in X} b(s_t \mid h_t) \prod_{k=t}^{j-1} K(z_{k+1}, s_{k+1} \mid \pi_k, s_k) = 1 \tag{60}$$

therefore:

$$\Pr(X^{\text{out}}) = 1 - \Pr(X^{\text{in}}),$$

and the bound is defined as follows:

$$R_{\max} \left(1 - \sum_{z_{t+1:j} \in Z^{\text{in}}_{t+1:j}} \sum_{s_{t:j} \in S^{\text{in}}_{t:j}} b(s_t \mid h_t) \prod_{k=t}^{j-1} P_T(s_{k+1} \mid \pi_k, s_k)\, P_Z(z_{k+1} \mid s_{k+1}) \right) \tag{61}$$

to summarize we have derived the non-robust bound, and the bound is defined as:

$$
\begin{aligned}
&\left|U_j^\pi(h_t) - \widehat{U}_j^\pi(h_t)\right| \leq \\
&R_{\max}\left(\Delta_b + \sum_{k=t}^{j-1}\left(\Delta_T(h_k, a_k) + \Delta_Z(h_k, a_k)\right)\right) + \\
&R_{\max}\left(1 - \sum_{z_{t+1:j} \in Z^{\text{in}}_{t+1:j}} \sum_{s_{t:j} \in S^{\text{in}}_{t:j}} b(s_t \mid h_t) \prod_{k=t}^{j-1} P_T(s_{k+1} \mid \pi_k, s_k)\, P_Z(z_{k+1} \mid s_{k+1})\right)
\end{aligned}
\tag{62}
$$

now (62) is the bound for a single time step, for the full value function bound of the entire horizon we need to some accross the time steps, therefore we get the final bound as:

$$
\begin{aligned}
\left|V_t^\pi(h_t) - \widehat{V}_t^\pi(h_t)\right| \leq R_{\max} \sum_{j=t}^{t+H} \Bigg[&\left(\Delta_b + \sum_{k=t}^{j-1}\left(\Delta_T(h_k, a_k) + \Delta_Z(h_k, a_k)\right)\right) + \\
&\left(1 - \sum_{z_{t+1:j} \in Z^{\text{in}}_{t+1:j}} \sum_{s_{t:j} \in S^{\text{in}}_{t:j}} b(s_t \mid h_t) \prod_{k=t}^{j-1} P_T(s_{k+1} \mid \pi_k, s_k)\, P_Z(z_{k+1} \mid s_{k+1})\right)\Bigg]
\end{aligned}
\tag{63}
$$

important note: over the support we have access to the actual reward, and we can you $r(s_j, a_j)$ instead of R_{max} to get a tighter bound ...

[clarify briefly novelty/difference wrt Moran's formulation?]

5 bounds robust

we will now use the bounds we showed in the non-robust section, in order to derive the bounds for the robust case. we wish to bound the theoretical robust value with the simplified (projected of the support) value function.As in the non-robust case, we first work at the level of a single reward contribution corresponding to a fixed reward index j, and only afterwards sum the resulting bound over the full horizon. The main additional difficulty in the robust setting is that the value is no longer evaluated under a fixed model, but rather under the worst-case model in the admissible uncertainty set. Therefore, the robust bound must compare two minimization problems (the theoretical VS projected). For a fixed reward index $j \in \{t, \ldots, t+H\}$, we define the robust reward contribution under the full uncertainty sets by:

$$
U_j^{\pi,\text{rob}}(h_t) \;:=\; \min_{\theta_{t:j-1} \in \Theta_{t:j-1}} U_j^\pi(h_t;\, \theta_{t:j-1}) \tag{64}
$$

and under the projected uncertainty sets by:

$$
\widehat{U}_j^{\pi,\text{rob}}(h_t) \;:=\; \min_{\hat{\theta}_{t:j-1} \in \hat{\Theta}_{t:j-1}} \widehat{U}_j^\pi\left(h_t;\, \hat{\theta}_{t:j-1}\right) \tag{65}
$$

such that:

$$
\theta = \left\{\{P_Z \in \mathcal{P}_Z\}_{k=t}^{j-1}, \{P_T \in \mathcal{P}_T\}_{k=t}^{j-1}\right\}, \quad \hat{\theta} = \left\{\{P_Z^{\text{in}} \in \widehat{\mathcal{P}}_Z\}_{k=t}^{j-1}, \{P_T^{\text{in}} \in \widehat{\mathcal{P}}_T\}_{k=t}^{j-1}\right\} \tag{66}
$$

where $\Theta_{t:j-1}, \hat{\Theta}_{t:j-1}$ denote the sets of all admissible full and projected model sequences, respectively. [this assumes rectangularity][is this definition for $\Theta_{t:j-1}$?] the first step in our robust derivation is to use a reduction lemma to connect the non-robust derivation to the robust version. first we formally write the robust difference of the theoretical with the projected:

$$\left|U_j^{\pi,\mathrm{rob}}(h_t) - \widehat{U}_j^{\pi,\mathrm{rob}}(h_t)\right| = \Bigg|\min_{\theta_{t:j-1}\in\Theta_{t:j-1}} \sum_{z_{t+1:j}} \sum_{s_{t:j}} \prod_{k=t}^{j-1} P_{T,k}(s_{k+1} \mid \pi_k, s_k)\ P_{Z,k}(z_{k+1} \mid s_{k+1})\ b(s_t \mid h_t)\, r(s_j, a_j)$$

$$- \min_{\hat{\theta}_{t:j-1}\in\widehat{\Theta}_{t:j-1}} \sum_{z_{t+1:j}\in Z^{\mathrm{in}}_{t+1:j}} \sum_{s_{t:j}\in S^{\mathrm{in}}_{t:j}} \prod_{k=t}^{j-1} P^{\mathrm{in}}_{T,k}(s_{k+1} \mid \pi_k, s_k)\ P^{\mathrm{in}}_{Z,k}(z_{k+1} \mid s_{k+1})\ \widehat{b}(s_t \mid h_t)\, r(s_j, a_j)\Bigg| \tag{67}$$

We use the fact that:

$$\left|\min_\theta f(\theta) - \min_{\hat\theta} \hat f(\hat\theta)\right| \le \max_{\theta,\hat\theta} \left|f(\theta) - \hat f(\hat\theta)\right| \tag{68}$$

proof:

Let

$$a := \min_\theta f(\theta), \qquad b := \min_{\hat\theta} \hat f(\hat\theta)$$

Then for any $\theta, \hat\theta$,

$$a \le f(\theta), \quad b \le \hat f(\hat\theta), \qquad \Rightarrow \qquad a - b \le f(\theta) - \hat f(\hat\theta)$$

Since this holds for every $\theta, \hat\theta$, we get

$$a - b \le \sup_{\theta,\hat\theta}\big(f(\theta) - \hat f(\hat\theta)\big) \le \sup_{\theta,\hat\theta}\big|f(\theta) - \hat f(\hat\theta)\big|$$

By swapping the roles of f and $\hat f$, we also obtain

$$b - a \le \sup_{\theta,\hat\theta}\big|f(\theta) - \hat f(\hat\theta)\big|$$

Therefore,

$$|a - b| \le \sup_{\theta,\hat\theta}\big|f(\theta) - \hat f(\hat\theta)\big|$$

and we obtained the following reduction lemma:

$$\left|U_j^{\pi,\mathrm{rob}}(h_t) - \widehat{U}_j^{\pi,\mathrm{rob}}(h_t)\right| \le \sup_{\substack{\theta_{t:j-1}\in\Theta_{t:j-1},\\ \hat\theta_{t:j-1}\in\widehat{\Theta}_{t:j-1}}} \left|U_j^{\pi}(h_t;\theta_{t:j-1}) - \widehat{U}_j^{\pi}(h_t;\hat\theta_{t:j-1})\right| \tag{69}$$

This inequality is the key bridge between the robust and non-robust analyses. The right-hand side no longer contains nested minimization problems; instead, it asks for a worst-case comparison between one admissible full-model sequence and one admissible projected-model sequence.

similarly to the non-robust bound, we use the triangle inequality and split the bound into 2 terms:

$$\le \sup_{\theta,\hat\theta} |\text{inside mismatch}| + \sup_{\theta} |\text{omitted trajectories contribution}| \tag{70}$$

for the first term (inside mismatch) we can use the same bound that we got in section 3, with a small difference. this time we will keep it at the kernel level and will not split it into transition and observation models like we did in (43). there are several reasons why in the robust case we decided to use the kernel level and not the split version. first it is tighter since:

$$\left\| K - \widehat{K} \right\|_1 \leq \text{transition part} + \text{observation part}$$

second, when we use:

$$\Delta_K^{rob}(h_k, a_k) \ := \ \sup_{\theta_k, \hat{\theta}_k} \left\| K^{\theta_k} - \widehat{K}^{\hat{\theta}_k} \right\|_1 \tag{71}$$

$\theta_k, \hat{\theta}_k$ are the one-step admissible full and projected model choices at stage k. it directly measures worst-case one-step mismatch between the full and projected trajectory kernels. It avoids having to decide whether the robust observation term should be weighted by P_T or P_T^{in}, by the smaller of the two, or by a supremum. All of that disappears if the theorem is stated at kernel level. the bound for the first term is then:

$$\sup_{\theta, \hat{\theta}} |\text{inside mismatch}| \ \leq \ R_{\max} \left(\Delta_b + \sum_{k=t}^{j-1} \Delta_K^{rob}(h_k, a_k) \right) \tag{72}$$

for the second term, the same omitted-trajectories argument from Section 3 applies. The only difference is that we now take the worst case over the admissible full-model sequence:

$$\sup_{\theta} |\text{omitted trajectories contribution}| \leq$$
$$R_{\max} \left(1 - \inf_{\theta_{t:j-1} \in \Theta_{t:j-1}} \sum_{z_{t+1:j} \in Z_{t+1:j}^{\text{in}}} \sum_{s_{t:j} \in S_{t:j}^{\text{in}}} b(s_t \mid h_t) \prod_{k=t}^{j-1} P_{T,k}(s_{k+1} \mid \pi_k, s_k)\, P_{Z,k}(z_{k+1} \mid s_{k+1}) \right) \tag{73}$$

Combining both terms, and summing over $j = t, \ldots, t+H$, we obtain the robust bound:

$$\begin{aligned} \left| V_t^{\pi,\text{rob}}(h_t) - \widehat{V}_t^{\pi,\text{rob}}(h_t) \right| \leq R_{\max} \sum_{j=t}^{t+H} \Bigg[& \left(\Delta_b + \sum_{k=t}^{j-1} \Delta_K^{rob}(h_k, a_k) \right) + \\ & \left(1 - \inf_{\theta_{t:j-1} \in \Theta_{t:j-1}} \sum_{z_{t+1:j} \in Z_{t+1:j}^{\text{in}}} \sum_{s_{t:j} \in S_{t:j}^{\text{in}}} b(s_t \mid h_t) \prod_{k=t}^{j-1} P_{T,k}(s_{k+1} \mid \pi_k, s_k)\, P_{Z,k}(z_{k+1} \mid s_{k+1}) \right) \Bigg] \end{aligned} \tag{74}$$